

Business Problem Statement

A leading commercial lending institution wants to optimize its credit risk management strategies, track capital liquidity, and better understand borrower behavior across its loan portfolio. The executive management team has noticed growing complexities in monitoring funded capital and identifying early default indicators across different risk tiers, geographic regions, and loan purposes. They are particularly interested in uncovering which structural factors—such as asset ownership, loan intent, and credit grade distribution—drive borrower performance, volume concentration, and capital recovery.

You are tasked with analysing the institution's commercial lending transactions dataset to answer the following overarching business question :—"How can the financial institution leverage transactional loan data to establish a centralized source of truth, mitigate credit risk exposure, and optimize portfolio liquidity management?"

Deliverables

1. Data Preparation & Modelling (Python): Clean and programmatically transform the raw, unformatted transaction logs. Address missing attributes, handle data type anomalies, resolve structural inconsistencies, and ensure full data integrity before database ingestion.
2. Data Analysis (SQL): Ingest and organize the structured data within a relational database instance (PostgreSQL). Write optimized queries leveraging analytical aggregations and joins to pre-calculate executive-level metrics, asset distributions, and volume concentrations.
3. Visualization & Insights (Power BI): Build a high-contrast, interactive executive dashboard utilizing a clean semantic star schema design. Integrate advanced analytics features (such as an AI Decomposition Tree) to enable stakeholders to seamlessly execute multi-dimensional, self-service data drilling.
4. Report and Presentation: Compile complex insights into a clean, story-based presentation designed for technical and non-technical stakeholders. Highlight key metric tracking, portfolio demand drivers, and data-driven risk management recommendations.
5. GitHub Repository: Structure and deploy all data cleaning scripts, database aggregation files, presentation documents, and the production Power BI file into a polished, professional portfolio repository.